Weekly report

Natalia Trigo Tomasevich

October 26, 2026

This week

- Competition model with p_0

This week

- Competition model with p_0
- Oblivious equilibrium

This week

- Competition model with p_0
- Oblivious equilibrium
- Simulations

Elements of the model

- Two symmetric providers, indexed by $j \in \{A, B\}$
- Two products, indexed by $i \in \{1, 2\}$
- Success probabilities: $0 < p_1 < p_2 < 1$
- Costs: $c_1 < c_2$
- Revenue: $R > c_2$
- p_0

Customer's beliefs

For the competition model, the customer observes the state x_A, x_B, where x_j is the fraction of successes over the interactions with provider j. The customer has a belied about each provider j

$$\theta_A \mid x_A \sim \text{Beta}(1 + \frac{x_A}{1-\rho}, 1 + \frac{1-x_A}{1-\rho}), \tag{1}$$

$$\theta_B \mid x_B \sim \text{Beta}(1 + \frac{x_B}{1-\rho}, 1 + \frac{1-x_B}{1-\rho}) \tag{2}$$

Customer's choice

The customer uses Thompson sampling (TS), drawing independently from the two belief distributions and choosing

$$y_t = \begin{cases} A, & \widetilde{\theta}_A > \max\{\widetilde{\theta}_B, p_0\}, \\ B, & \widetilde{\theta}_B > \max\{\widetilde{\theta}_A, p_0\}, \\ 0, & p_0 > \max\{\widetilde{\theta}_A, \widetilde{\theta}_B\}. \end{cases} \tag{3}$$

Here 0 denotes the outside option of fixed quality p_0. Ties have probability zero. The selection probabilities are

$$\begin{aligned} D_A(x_A, x_B; p_0) &= \mathbb{P}(\widetilde{\theta}_A > \max\{\widetilde{\theta}_B, p_0\} \mid x_A, x_B), \\ D_B(x_A, x_B; p_0) &= \mathbb{P}(\widetilde{\theta}_B > \max\{\widetilde{\theta}_A, p_0\} \mid x_A, x_B), \\ D_0(x_A, x_B; p_0) &= \mathbb{P}(\widetilde{\theta}_A < p_0 \mid x_A)\mathbb{P}(\widetilde{\theta}_B < p_0 \mid x_B). \end{aligned} \tag{4}$$

Hence $D_A + D_B + D_0 = 1$.

State transitions

Superscripts s and f denote success and failure, respectively. The state transitions are

$$\begin{aligned} T_A^s(x) &= (\rho x_A + 1 - \rho, x_B), \\ T_A^f(x) &= (\rho x_A, x_B), \\ T_B^s(x) &= (x_A, \rho x_B + 1 - \rho), \\ T_B^f(x) &= (x_A, \rho x_B). \end{aligned} \tag{5}$$

Strategies

We consider stationary mixed strategies that depend only on the seller's own reputation. For each seller $j \in \{A, B\}$, a strategy is a mapping

$$\sigma_j : [0, 1] \to \Delta(\{1, 2\}), \tag{6}$$

where $\sigma_j(i \mid x_j)$ denotes the probability that seller j chooses product i when selected at reputation x_j.

Distribution-Based Oblivious Equilibrium

For fixed p_0, seller A conjectures a stationary distribution μ_B for its competitor's reputation before customer selection. Its expected demand is

$$d_A(x_A; \mu_B, p_0) = \int_0^1 D_A(x_A, y; p_0)\, \mu_B(dy). \tag{7}$$

The seller's problem becomes

$$\begin{aligned} U_A(x_A) = \gamma[1 - d_A(x_A; \mu_B, p_0)]U_A(x_A) \\ + d_A(x_A; \mu_B, p_0) \max_{i \in \{1,2\}} \Big\{ R - c_i \\ + \gamma \left[p_i U_A(T_\rho^s(x_A)) + (1 - p_i) U_A(T_\rho^f(x_A)) \right] \Big\}. \end{aligned} \tag{8}$$

The term $1 - d_A$ includes choosing B or the outside option.

Distribution-Based Oblivious Equilibrium

- Given the conjectured distribution μ_B, each seller solves its Bellman equation and obtains an optimal strategy σ.
- We require this distribution to be consistent with the reputations generated when both sellers follow σ.

Joint transition probability kernel

For fixed p_0, define the joint transition kernel (with p_0 implicit in P^σ) as

$$P^\sigma(x, C) = \mathbb{P}(x_{t+1} \in C \mid x_t = x, \sigma),$$

which is equivalent to

$$\begin{aligned} P^\sigma(x, C) = {} & D_0(x_A, x_B; p_0)\mathbf{1}\{x \in C\} \\ & + D_A(x_A, x_B; p_0)\Big[p_{\sigma_A}(x_A)\mathbf{1}\{T_A^s(x) \in C\} \\ & \quad + (1 - p_{\sigma_A}(x_A))\mathbf{1}\{T_A^f(x) \in C\}\Big] \\ & + D_B(x_A, x_B; p_0)\Big[p_{\sigma_B}(x_B)\mathbf{1}\{T_B^s(x) \in C\} \\ & \quad + (1 - p_{\sigma_B}(x_B))\mathbf{1}\{T_B^f(x) \in C\}\Big]. \end{aligned} \tag{9}$$

If the outside option is chosen, both reputations remain unchanged.

Distribution-Based Oblivious Equilibrium

A symmetric oblivious equilibrium is a triple (σ^*, μ^*, ν^*) satisfying:

1. Optimality

$$\operatorname{supp} \sigma^*(\cdot \mid x) \subseteq \arg \max_{i \in \{1,2\}} \Big\{ R - c_i + \gamma \big[p_i U_{\mu^*}(T_\rho^s(x)) + (1 - p_i) U_{\mu^*}(T_\rho^f(x)) \big] \Big\}, \quad \forall x \in [0,1].$$

2. Consistency

$$\nu^* = \nu^* P^{\sigma^*},$$
$$\mu^*(C) = \nu^*(C \times [0,1]) = \nu^*([0,1] \times C), \quad \forall C \subseteq [0,1] \text{ measurable.}$$

What did we find in this model

- For small p_0, in particular $p_0 < p_1$, we have that the fraction of use of product 2 among the sales of a seller is larger than the monopoly seller
- For larger p_0, for example $p_0 = p_2$ we find a case where this fraction is larger for the monopoly
- Why?

Simulations

go to vscode, there are too many plots